Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 16: Approved Solutions

Equivalence Relations

Covers Chartrand & Zhang, Section 5.2

Status of this set

Ungraded practice. Solutions released with it. Write the proofs in *complete sentences* — that is the skill being practised.

Part I: How to Practice This Material

Three proofs, clearly labelled

The template

Reflexive. Let $a \in A$. . . . Therefore $a \, R \, a$.
Symmetric. Assume $a \, R \, b$. . . . Therefore $b \, R \, a$.
Transitive. Assume $a \, R \, b$ and $b \, R \, c$. . . . Therefore $a \, R \, c$.
Conclude. Since R is reflexive, symmetric, and transitive, it is an equivalence relation.

The recurring move

In the transitive part you are handed two facts about a, b and b, c and asked for a fact about a, c. **Combine the hypotheses — usually by adding — and let b cancel.**
That single move handles most parity and divisibility relations you will meet.

Three mistakes to expect

1. **Proving two properties and stopping.** All three are required.
2. **Assuming what is to be proved.** In the symmetric part you may use $a \, R \, b$; you may not use $b \, R \, a$.
3. **Writing "$2k$" without saying k is an integer.** The definition of even requires it, and the last line of every parity proof depends on it.

Part II: Practice Problems

Section A — Is it an equivalence relation?

1. For each relation on $\mathbb{Z}$, decide whether it is an equivalence relation. Prove it or name the failing property with a witness.

(a) $a \mathrel{R} b$ if $a - b$ is divisible by 5.

(b) $a \mathrel{R} b$ if $a \cdot b$ is even.

(c) $a \mathrel{R} b$ if $a^2 = b^2$.

Approved solution

(a) Yes. *Reflexive:* $a - a = 0 = 5 \cdot 0$. *Symmetric:* if $a - b = 5k$ then $b - a = 5(-k)$, and $-k$ is an integer. *Transitive:* if $a - b = 5k$ and $b - c = 5m$, then $a - c = (a - b) + (b - c) = 5(k + m)$, and $k + m$ is an integer.
(b) No. Not reflexive: take $a = 3$. Then $a \cdot a = 9$ is odd, so $3 \not R\, 3$. (It is also not transitive: $3 \mathrel{R} 2$ and $2 \mathrel{R} 5$, but $3 \cdot 5 = 15$ is odd.)
(c) Yes. *Reflexive:* $a^2 = a^2$. *Symmetric:* if $a^2 = b^2$ then $b^2 = a^2$. *Transitive:* if $a^2 = b^2$ and $b^2 = c^2$ then $a^2 = c^2$.
All three follow from the corresponding properties of equality. Note $a^2 = b^2$ means $|a| = |b|$, so this is the same relation as Chartrand's Supplementary Exercise 14.

Part (b) is worth care. Many students check symmetry, see that it holds, and stop. One failing property is enough — but you have to look for it.

Section B — Full proofs

2. Prove that the relation R on $\mathbb{Z}$ defined by $a \mathrel{R} b$ if $a + 3b$ is even is an equivalence relation.

Approved solution

Reflexive. Let $a \in \mathbb{Z}$. Then $a + 3a = 4a = 2(2a)$, and $2a$ is an integer, so $a + 3a$ is even. Hence $a \mathrel{R} a$.
Symmetric. Assume $a \mathrel{R} b$, so $a + 3b$ is even, say $a + 3b = 2k$ with $k \in \mathbb{Z}$. We must show $b + 3a$ is even. Observe

$$(a + 3b) + (b + 3a) = 4a + 4b = 2(2a + 2b),$$

so $b + 3a = 2(2a + 2b) - 2k = 2(2a + 2b - k)$. Since $2a + 2b - k$ is an integer, $b + 3a$ is even and $b \mathrel{R} a$.
Transitive. Assume $a \mathrel{R} b$ and $b \mathrel{R} c$, so $a + 3b = 2k$ and $b + 3c = 2m$ for integers k, m. Then

$$(a + 3b) + (b + 3c) = a + 4b + 3c = 2k + 2m,$$

so $a + 3c = 2k + 2m - 4b = 2(k + m - 2b)$. Since $k + m - 2b$ is an integer, $a + 3c$ is even and $a \mathrel{R} c$. ■

Both the symmetric and transitive parts used the same trick: add the expression you have to the one you want, and check the sum is visibly even. Then subtract. Worth naming as a technique, because it recurs on every relation of this form.

3. Let S be a nonempty set and let R_1 and R_2 be equivalence relations on S. Prove that $R_1 \cap R_2$ is an equivalence relation on S.

Approved solution

Write $a\ R\ b$ for $(a,b) \in R_1 \cap R_2$, which means $a\ R_1\ b$ *and* $a\ R_2\ b$.
Reflexive. Let $a \in S$. Since R_1 and R_2 are each reflexive, $(a,a) \in R_1$ and $(a,a) \in R_2$, so $(a,a) \in R_1 \cap R_2$.
Symmetric. Assume $(a,b) \in R_1 \cap R_2$. Then $(a,b) \in R_1$, and R_1 is symmetric, so $(b,a) \in R_1$. Likewise $(b,a) \in R_2$. Hence $(b,a) \in R_1 \cap R_2$.
Transitive. Assume (a,b) and (b,c) both lie in $R_1 \cap R_2$. Then (a,b) and (b,c) are in R_1, so $(a,c) \in R_1$ by transitivity of R_1; the same argument in R_2 gives $(a,c) \in R_2$. Hence $(a,c) \in R_1 \cap R_2$. ■

Notice the shape. Each part does the same thing twice, once in each relation. Because $\cap$ is a conjunction, everything that holds in both components holds in the intersection.
The union does not behave this way — see the challenge problem.

Section C — Recognising them

4. Which of these are equivalence relations? One sentence each.

(a) "Has the same birthday as," on the set of people.

(b) "Is a sibling of," on the set of people.

(c) "Has the same file extension as," on a set of filenames.

(d) "Is logically equivalent to," on compound statements.

Approved solution

(a) Yes. Same-birthday is reflexive, symmetric, and transitive because equality of dates is. The classes are the 366 possible birthdays.
(b) No. Not reflexive under the usual meaning — a person is not their own sibling. (It is also arguably not transitive, depending on whether half-siblings count.)
(c) Yes. The classes are the groups of files sharing an extension. This is exactly what a file manager does when it sorts by type.
(d) Yes. $\equiv$ is reflexive, symmetric, and transitive — which is why you could chain rewrites in Week 1 and still be discussing the same statement.

Pattern worth naming: (a), (c), and (d) all say "f of one equals f of the other" for some function f — birthday, extension, truth column. **Any relation of that form is automatically an equivalence relation**, because equality is.

Section D — Writing

5. In five to eight complete sentences, explain why proving a relation is an equivalence relation requires three separate arguments, and why showing it is *not* one requires only one. Connect this to something you proved earlier in the term.

Approved solution

Model answer. By Definition 5.14, being an equivalence relation means being reflexive *and* symmetric *and* transitive, and a conjunction of three statements is true only when all three parts are true. So establishing it requires establishing each part separately, and demonstrating two of them leaves the conjunction unsettled. Showing a relation is not an equivalence relation is the negation of that conjunction, and by De Morgan the negation of "all three hold" is "at least one fails," so exhibiting a single failing property is enough. This is the same structure as proving a biconditional in Lecture 6, where $P \Leftrightarrow Q$ is a conjunction of two implications and both had to be proved. It is also why one counterexample suffices: each property is a universal statement, so its negation is existential and a witness settles it.

Full credit requires identifying the definition as a *conjunction* and applying De Morgan to its negation. An answer that only describes the procedure earns about half.

Section E — Challenge (optional)

6. Show that the union of two equivalence relations need not be an equivalence relation. Give a specific example and say exactly which property fails.

Approved solution

Let $S = \{1, 2, 3\}$ and take

$$R_1 = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}, \qquad R_2 = \{(1,1),(2,2),(3,3),(2,3),(3,2)\}.$$

Each is an equivalence relation: R_1 has classes $\{1,2\}$ and $\{3\}$, and R_2 has classes $\{1\}$ and $\{2,3\}$.
Their union is $R_1 \cup R_2 = \{(1,1),(2,2),(3,3),(1,2),(2,1),(2,3),(3,2)\}$.
This is reflexive and symmetric, but **not transitive**: $(1,2)$ and $(2,3)$ are both present while $(1,3)$ is not.

Why the union misbehaves and the intersection does not. Transitivity requires a *closure* property — given two links, a third must exist. Intersecting two relations keeps only pairs both already have, so any chain in the intersection is a chain in each, and each supplies the shortcut. Union introduces chains that neither original relation contained, with one link from each, and neither is obliged to close it.
In partition language: R_1 groups 1 with 2, and R_2 groups 2 with 3. The union tries to hold both groupings at once, which forces 1 and 3 together — and that pair was never added.

Instructor notes

- **Problem 2 teaches the add-and-subtract technique** used in both the symmetric and transitive parts. It generalises to every "$\alpha a + \beta b$ is even" relation, which is what Chartrand's Examples 5.21 and 5.25 and Supplementary Exercise 12 all are.
- **Problem 4's closing observation** — any relation of the form "$f(a) = f(b)$" is automatically an equivalence relation — is worth stating in class. It explains same-birthday, same-extension, and congruence in one stroke.
- **Problem 6** is the natural companion to Problem 3. Intersection works, union does not, and the reason is that transitivity is a closure condition. Put both on the board together.
- **Problem 1(b)** catches students who check one property and stop.

- Reserved for Homework 7: Section 5.2 Exercises 2, 4, 6, 12, 14, 16 (even). Odd ones have printed answers and are self-check only.